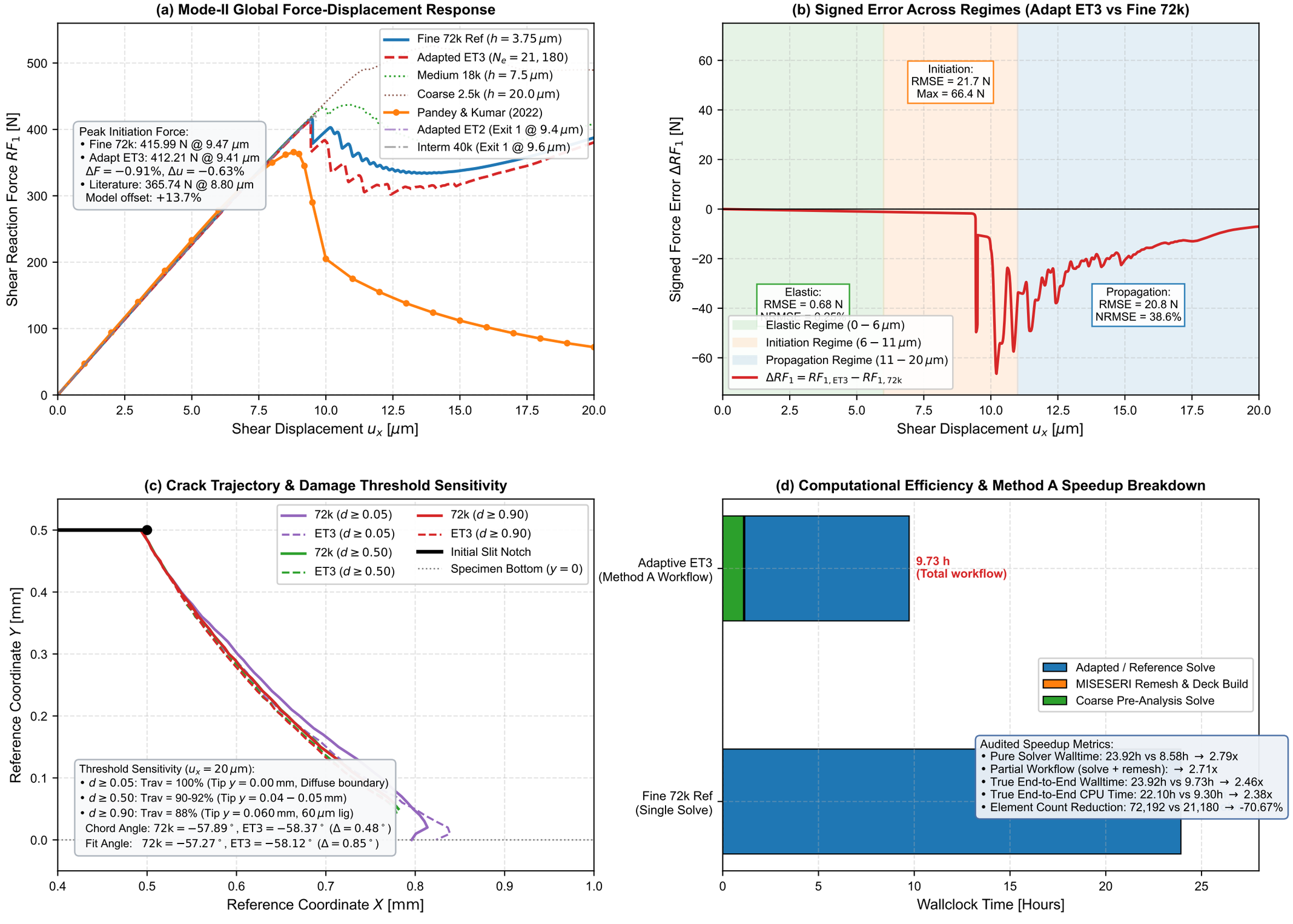

(a) Mode-II Global Force-Displacement Response
Fine 72k Ref ($h = 3.75\,\mu$m)
Adapted ET3 ($N_e = 21,180$)
Medium 18k ($h = 7.5\,\mu$m)
Coarse 2.5k ($h = 20.0\,\mu$m)
Pandey & Kumar (2022)
Adapted ET2 (Exit 1 @ 9.4 μm)
Interm 40k (Exit 1 @ 9.6 μm)
Peak Initiation Force:
• Fine 72k: 415.99 N @ 9.47 μm
• Adapt ET3: 412.21 N @ 9.41 μm
$\Delta F = -0.91\%$, $\Delta u = -0.63\%$
• Literature: 365.74 N @ 8.80 μm
Model offset: +13.7%
Shear Reaction Force RF_1 [N]
Shear Displacement u_x [μm]
(b) Signed Error Across Regimes (Adapt ET3 vs Fine 72k)
Initiation:
RMSE = 21.7 N
Max = 66.4 N
Elastic:
RMSE = 0.68 N
Propagation:
RMSE = 20.8 N
NRMSE = 38.6%
Elastic Regime ($0 - 6\,\mu$m)
Initiation Regime ($6 - 11\,\mu$m)
Propagation Regime ($11 - 20\,\mu$m)
$\Delta RF_1 = RF_{1,\mathrm{ET3}} - RF_{1,\mathrm{72k}}$
Signed Force Error ΔRF_1 [N]
Shear Displacement u_x [μm]
(c) Crack Trajectory & Damage Threshold Sensitivity
72k ($d \geq 0.05$)
ET3 ($d \geq 0.05$)
72k ($d \geq 0.50$)
ET3 ($d \geq 0.50$)
72k ($d \geq 0.90$)
ET3 ($d \geq 0.90$)
Initial Slit Notch
Specimen Bottom ($y = 0$)
Threshold Sensitivity ($u_x = 20\,\mu$m):
• $d \geq 0.05$: Trav = 100% (Tip $y = 0.00$ mm, Diffuse boundary)
• $d \geq 0.50$: Trav = 90-92% (Tip $y = 0.04 - 0.05$ mm)
• $d \geq 0.90$: Trav = 88% (Tip $y = 0.060$ mm, 60 μm lig)
Chord Angle: 72k = −57.89°, ET3 = −58.37° ($\Delta = 0.48°$)
Fit Angle: 72k = −57.27°, ET3 = −58.12° ($\Delta = 0.85°$)
Reference Coordinate Y [mm]
Reference Coordinate X [mm]
(d) Computational Efficiency & Method A Speedup Breakdown
Adaptive ET3
(Method A Workflow)
9.73 h
(Total workflow)
Fine 72k Ref
(Single Solve)
Adapted / Reference Solve
MISESERI Remesh & Deck Build
Coarse Pre-Analysis Solve
Audited Speedup Metrics:
• Pure Solver Walltime: 23.92h vs 8.58h → 2.79x
• Partial Workflow (solve + remesh): → 2.71x
• True End-to-End Walltime: 23.92h vs 9.73h → 2.46x
• True End-to-End CPU Time: 22.10h vs 9.30h → 2.38x
• Element Count Reduction: 72,192 vs 21,180 → -70.67%
Wallclock Time [Hours]